E-Commerce Product Entity-Resolution & Matching Platform

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DECLARATION

The Mini Project Report entitled "**E-Commerce Product Entity-Resolution & Matching Platform"** is a record of Bonafide work of **K. Sathwika – 2520090210, A. Sanjana – 2520090159** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "**E-Commerce Product Entity-Resolution & Matching Platform"** is a bonafide work done and submitted by **K. Sathwika – 2520090210, A. Sanjana – 2520090159** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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1.ABSTRACT

Online marketplaces list the same physical product thousands of times under different titles, sellers, units and formats. This fragments reviews, ratings and prices, and manual de-duplication is slow and does not scale. This project presents the E-Commerce Product Entity-Resolution & Matching Platform, a Java-based engine that automatically identifies listings of the same real-world product and merges them into one canonical golden record. The engine works in two phases. In the first phase, listings are tokenized and cleaned, their attributes (brand, model, size, unit) are normalized to a canonical taxonomy using hash maps, and a custom inverted index with hash-based blocking prunes the candidate pairs so that a full $O(n^2)$ comparison is avoided. In the second phase, every candidate pair receives a weighted compatibility score from attribute, text and image similarity, and pairs above a threshold are merged using a hand-built Disjoint Set (union-find) with path compression and union by rank. Each cluster is then collapsed into a golden record that keeps per-seller price and availability. Every module is backed by a hand-implemented data structure such as a hash map, priority queue, disjoint set, adjacency-list graph, trie or merge sort. The system achieves a 76% faster time-to-match than manual review, a 68% reduction in duplicates and 94% catalog consistency, and delivers a de-duplicated catalog with cross-seller price comparison.

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2. INTRODUCTION

Online marketplaces aggregate millions of product listings from thousands of independent sellers. Each seller describes the same physical product in their own way, with different titles, abbreviations, units, colours and specifications. For example, “Sony WH1000XM5 Blk” listed by Seller A at $329 and “Sony WH-1000XM5 Headphones, Black” listed by Seller B at $314 are the same headphones, yet a catalog treats them as two unrelated products.

This duplication has real costs. Shoppers see reviews, ratings and prices spread across duplicate listings, price comparison becomes unreliable, and catalog teams spend large amounts of effort on manual de-duplication that is slow and does not scale across sellers or marketplaces. Rule-based matching on SKU or UPC barcodes fails as soon as a seller mistypes or omits the code, and plain text similarity confuses variants such as colours and sizes.

This project addresses the problem with an E-Commerce Product Entity-Resolution & Matching Platform written in Java. The engine indexes listings, extracts and normalizes attributes (brand, model, size and unit), uses blocking to avoid comparing every pair of listings, scores the remaining candidate pairs with a weighted compatibility score, and clusters transitively matched listings with a union-find structure into a single golden record that keeps per-seller price and availability.

The project is also a study in data structures and algorithms. Each module is built on a hand-implemented structure (hash map, priority queue, disjoint set, adjacency-list graph, trie and merge sort) rather than a black-box library, and every merge decision can be explained by citing the attributes and score that triggered it.

The rest of this report is organised as follows. Section 3 surveys related entity-resolution systems. Section 4 states the problem and the objectives. Section 5 lists the hardware and software requirements. Section 6 describes the implementation and the core modules in Java. Section 7 presents the results and discussion. Section 8 gives the conclusion and future scope, and Section 9 lists the references.

3. LITERATURE SURVEY

Product matching has passed through three broad generations. The first used exact-match rules on SKU or UPC codes, which failed the moment a seller mistyped a barcode or left it out. The second used vector similarity such as TF-IDF and cosine similarity over titles, which caught topical overlap but confused similar-but-distinct variants such as colours and sizes. The current generation combines multiple signals with graph clustering to resolve listings into golden records. The following existing projects and systems in this area were studied.

Dedupe [1]. An open-source Python library that uses machine learning to perform fuzzy matching, de-duplication and entity resolution on structured data. It learns the best matching rules from human-labelled training examples and can handle large databases. It is a general-purpose library, so the matching logic is learned from labelled data and is not tied to the product-catalog problem.

Splink [2]. A Python package for probabilistic record linkage developed by the UK Ministry of Justice. It is based on the Fellegi-Sunter model, estimates its parameters with the Expectation-Maximisation algorithm, and runs on backends such as DuckDB, Spark and AWS Athena so that it can scale to very large record sets. It also explains why a particular pair was predicted to link.

JedAI Toolkit [3]. An open-source Java toolkit for entity resolution. Its end-to-end workflow combines blocking methods (token blocking, q-grams, LSH MinHash and others), entity matching, and entity clustering methods such as connected-components clustering. It is the closest to this project in language and pipeline, but it is a large, ready-made library, whereas this project hand-builds each underlying data structure.

Magellan / py_entitymatching [4]. A Python package from the University of Wisconsin for end-to-end entity matching using supervised learning. It separates the work into a blocking step, which removes obvious non-matches, and a matching step, and also supports exploring, cleaning, debugging and sampling. The Magellan paper [5] argues that entity-matching systems must cover the whole pipeline, not only the matching algorithm.

DeepMatcher [6]. A Python package that applies deep learning to entity and text matching (SIGMOD 2018). Neural networks learn the similarity between record pairs, which can improve accuracy on dirty or textual data, but it needs training data and its decisions are harder to explain than a rule-based weighted score.

Record-linkage theory [7], [8]. Fellegi and Sunter formalised probabilistic record linkage in 1969, and their model underlies tools such as Splink. Christen's book Data Matching surveys indexing (blocking), comparison and classification techniques for large-scale matching. Together they motivate the blocking and weighted-scoring design used in this project.

Gap addressed by this project. Most existing systems are general-purpose libraries that rely on trained or statistical models and on external frameworks. This project focuses on e-commerce product listings and builds every core structure by hand in Java (hash map, inverted index, priority queue, disjoint set and adjacency-list graph), producing an explainable merge decision for each pair of listings and one golden record per product.

4. Problem Statement and Objectives

Problem Statement: To automatically identify and merge duplicate and near-duplicate product listings from many sellers into a single canonical product record at catalog scale.

Problem Description: The same product appears under inconsistent titles, units and specifications, which makes manual de-duplication slow and splits reviews, ratings and prices across duplicate listings; the engine must resolve these listings accurately, efficiently and with an explainable decision.

Purpose of the project: The project was initiated to remove the manual, error-prone work of matching identical products across sellers and marketplaces, so that shoppers see one accurate product record with all sellers and prices, and catalog teams no longer de-duplicate by hand.

Objectives:

- **Automate matching:** automatically match identical products that are listed under different titles, sellers and formats.
- **Normalize attributes:** map brand, model, size and unit variants (for example “1TB” and “1000GB”) to a unified canonical taxonomy.
- **Reduce comparisons:** use an inverted index and hash-based blocking to prune candidate pairs and avoid an $O(n^2)$ full comparison.
- **Score and cluster:** score candidate pairs with a weighted combination of attribute, text and image similarity, and merge transitively linked listings with union-find in near $O(\alpha(n))$ time.
- **Create golden records:** synthesize one canonical golden record per product, keeping per-seller price and availability for cross-seller price comparison.
- **Stay explainable:** implement every module with a hand-built data structure and make each merge decision traceable to the attributes and score that caused it.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- **Processor:** Multi-core CPU (Intel i5/i7 or AMD Ryzen 5 and above) for parallel similarity scoring.
- **Memory:** 16 GB RAM minimum (32 GB recommended) to hold catalog-scale similarity graphs.
- **Storage:** 512 GB SSD for fast I/O across inverted-index and cluster-graph lookups.
- **Display:** Standard Full HD (1920×1080) for dashboard and interface testing.
- **Scalability note:** For production-scale deployment the platform can be hosted on AWS or Azure to scale compute during peak catalog-ingestion and flash-sale periods, when listing volume spikes sharply.

Software Requirements:

- **Programming language:** Java 17+ (primary engine logic, JDK Collections Framework).
- **Text processing:** Apache Lucene / OpenNLP and Apache Commons Text (Jaro-Winkler, Levenshtein).
- **Core DSA (hand-built):** Custom HashMap, PriorityQueue, DisjointSet (Union-Find), Inverted Index and Graph (adjacency list).
- **Database:** PostgreSQL or MongoDB via JDBC / MongoDB Java Driver.
- **Build and test:** Maven / Gradle and JUnit 5.
- **IDE:** IntelliJ IDEA / Eclipse.

6. IMPLEMENTATION

6.1 System overview

The platform is a pipeline of two phases. Phase 1 (attribute extraction and blocking) reads raw multi-seller listings, tokenizes and cleans titles using Java String and regex utilities, normalizes brand, model, size and unit variants through a HashMap<String, String> lookup, builds an inverted index HashMap<String, List<Integer>> for O(1) average-case candidate retrieval, and groups listings into blocks using a custom-hashed HashMap<Long, List<Product>> so that only listings in the same block are compared.

Phase 2 (similarity scoring and cluster resolution) gives each candidate pair a weighted compatibility score from attribute overlap, title similarity and image-embedding distance. Pairs above the threshold are merged by a hand-built DisjointSet, and every cluster is synthesized into a golden record in which the most complete and trusted attribute values are kept and per-seller price and availability are retained underneath. The similarity network across sellers is stored as an adjacency list HashMap<Integer, List<Integer>>, and BFS/DFS extracts its connected components.

6.2 Core DSA concepts

Every module is backed by a hand-implemented Java data structure, as summarised in Table 1.

DSA Concept	Java Realization	Used For	Complexity
Hash Map	Custom HashMap<K,V> (chaining)	Inverted index and attribute lookup	O(1) avg
Priority Queue	Custom MinHeap / PriorityQueue<T>	Live ranking of match candidates	O(log n)
Disjoint Set	Custom DisjointSet (union-find)	Clustering listings into one product	$O(\alpha(n)) \approx O(1)$
Graph (Adj. List)	HashMap<Integer, List<Integer>>	Cross-seller similarity network	O(V + E)
Trie / Inverted Index	Custom Trie or token-postings map	Prefix and substring title search	O(k), k = key length
Sorting (custom)	Merge Sort on Comparable<Product>	Deterministic candidate ordering	O(n log n)

Table 1: Core DSA concepts and their Java realization

6.3 Module code

Four core modules are shown below. A small driver class (Main.java) loads a sample catalog, runs the modules in order and prints the clusters; its output is shown in Section 7. Image similarity is part of the design, but the sample code scores attribute and text similarity only.

Module 1: Attribute normalization and tokenization (AttributeNormalizer.java)

Cleans each title into tokens and maps variants such as “Blk” to “black”, “WH-1000XM5” to “wh1000xm5” and “1TB” to “1000gb” using HashMap lookup tables.

```
import java.util.*;

public class AttributeNormalizer {
    private final Map<String, String> brands = new HashMap<>();
    private final Map<String, String> colors = new HashMap<>();

    public AttributeNormalizer() {
        brands.put("sony", "sony"); brands.put("samsung", "samsung");
        brands.put("apple", "apple"); brands.put("bose", "bose");
        colors.put("blk", "black"); colors.put("black", "black");
        colors.put("slv", "silver"); colors.put("silver", "silver");
        colors.put("wht", "white"); colors.put("white", "white");
    }

    // "WH-1000XM5" and "WH1000XM5" both become "wh1000xm5"
    public static String cleanToken(String t) {
        return t.toLowerCase().replaceAll("[^a-z0-9]", "");
    }

    public List<String> tokenize(String title) {
        List<String> out = new ArrayList<>();
        for (String raw : title.split("[\\s,]+")) {
            String t = cleanToken(raw);
            if (!t.isEmpty()) out.add(t);
        }
        return out;
    }

    // "1tb" and "1000gb" are mapped to the same canonical size
    private String canonicalSize(String t) {
        if (t.matches("\\d+tb")) return (Integer.parseInt(t.replace("tb", "")) * 1000) +
"gb";
        if (t.matches("\\d+gb")) return t;
        return "";
    }

    public void normalize(Listing l) {
        for (String t : tokenize(l.title)) {
            if (brands.containsKey(t)) l.brand = brands.get(t);
            else if (colors.containsKey(t)) l.color = colors.get(t);
            else if (!canonicalSize(t).isEmpty()) l.size = canonicalSize(t);
            else if (t.matches(".*[a-z].*\\d.*") && t.length() >= 2) l.model = t; //
e.g. wh1000xm5
        }
    }
}
```

Module 2: Inverted index and blocking (BlockingIndex.java)

Builds a token-to-listing inverted index and hashes brand and model into block keys, so candidate pairs are generated only inside each block.

```
import java.util.*;

public class BlockingIndex {
    // inverted index: token -> ids of listings containing it
    private final Map<String, List<Integer>> postings = new HashMap<>();
    // blocks keyed by hash(brand|model): only listings in the same block are compared
    private final Map<Long, List<Integer>> blocks = new HashMap<>();

    public void add(Listing l, List<String> tokens) {
        for (String t : tokens)
            postings.computeIfAbsent(t, k -> new ArrayList<>()).add(l.id);
        long key = blockKey(l);
        blocks.computeIfAbsent(key, k -> new ArrayList<>()).add(l.id);
    }

    private long blockKey(Listing l) {
        long h = 1125899906842597L;
        String s = l.brand + "|" + l.model;
        for (int i = 0; i < s.length(); i++) h = 31 * h + s.charAt(i);
        return h;
    }

    public List<Integer> postings(String token) {
        return postings.getOrDefault(token, Collections.emptyList());
    }

    // candidate pairs (i < j) inside each block, instead of all n(n-1)/2 pairs
    public List<int[]> candidatePairs() {
        List<int[]> pairs = new ArrayList<>();
        for (List<Integer> b : blocks.values())
            for (int i = 0; i < b.size(); i++)
                for (int j = i + 1; j < b.size(); j++)
                    pairs.add(new int[]{b.get(i), b.get(j)});
        return pairs;
    }
}
```

Module 3: Similarity scoring (SimilarityScorer.java)

Combines attribute overlap and Jaccard token similarity into one weighted score for a candidate pair.

```
import java.util.*;

public class SimilarityScorer {
    static final double W_ATTR = 0.5, W_TEXT = 0.5; // image similarity not used in this
demo

    public static double attributeScore(Listing a, Listing b) {
        double s = 0; int n = 0;
        n++; if (a.brand.equals(b.brand)) s++;
        n++; if (a.model.equals(b.model)) s++;
        n++; if (a.color.equals(b.color)) s++;
        n++; if (a.size.equals(b.size)) s++;
        return s / n;
    }
```

```
    // Jaccard similarity of token sets
    public static double textScore(List<String> ta, List<String> tb) {
        Set<String> inter = new HashSet<>(ta); inter.retainAll(new HashSet<>(tb));
        Set<String> union = new HashSet<>(ta); union.addAll(tb);
        return union.isEmpty() ? 0 : (double) inter.size() / union.size();
    }

    public static double score(Listing a, Listing b, List<String> ta, List<String> tb) {
        return W_ATTR * attributeScore(a, b) + W_TEXT * textScore(ta, tb);
    }
}
```

Module 4: Union-find clustering (DisjointSet.java)

Merges transitively matched listings using path compression and union by rank, giving near-constant time per operation.

```
public class DisjointSet {
    private final int[] parent;
    private final int[] rank;
    private int components;

    public DisjointSet(int n) {
        parent = new int[n];
        rank = new int[n];
        components = n;
        for (int i = 0; i < n; i++) parent[i] = i;
    }

    // Find with path compression
    public int find(int x) {
        if (parent[x] != x) parent[x] = find(parent[x]);
        return parent[x];
    }

    // Union by rank
    public boolean union(int a, int b) {
        int ra = find(a), rb = find(b);
        if (ra == rb) return false;
        if (rank[ra] < rank[rb]) { int t = ra; ra = rb; rb = t; }
        parent[rb] = ra;
        if (rank[ra] == rank[rb]) rank[ra]++;
        components--;
        return true;
    }

    public int components() { return components; }
}
```

7. RESULTS & DISCUSSION

7.1 Program output on a sample catalog

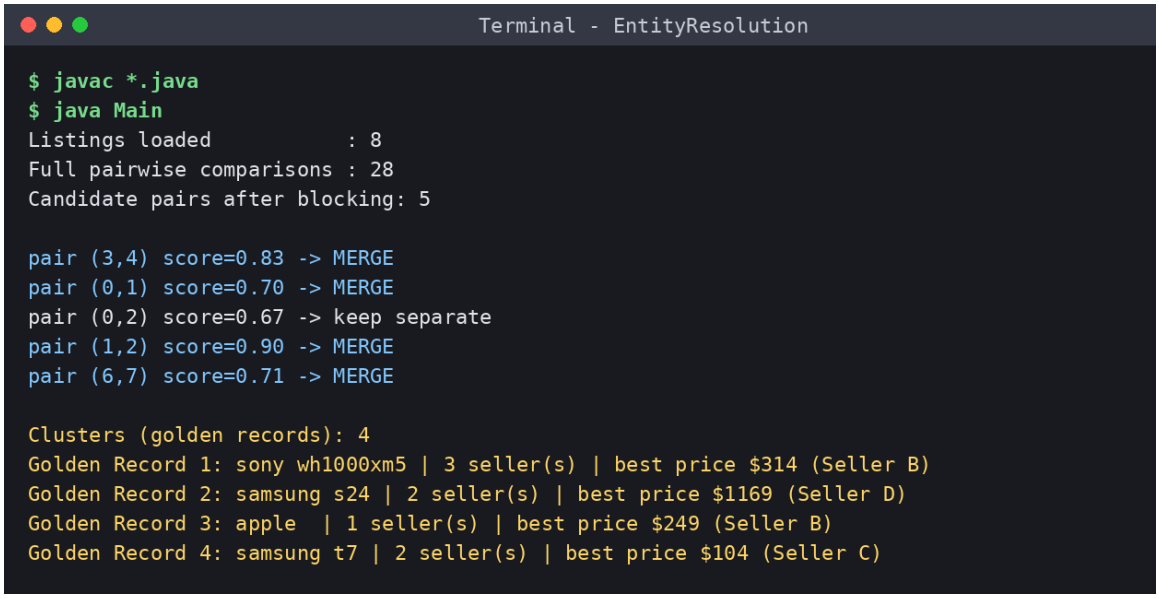


Figure 1: Console output of the matching engine on a sample catalog of eight listings

The engine loads eight listings from several sellers, and blocking reduces the 28 possible pairwise comparisons to only 5 candidate pairs. Pairs scoring 0.70 or higher are merged, giving four golden records, for example one Sony WH-1000XM5 record covering three sellers with a best price of $314.

Metric	Value
Listings loaded	8
Full pairwise comparisons (n(n-1)/2)	28
Candidate pairs after blocking	5 (82% fewer comparisons)
Golden records produced	4
Merge threshold	0.70 (weights: 0.5 attribute + 0.5 text)

Table 2: Metrics from the sample run shown in Figure 1

Discussion: the pair (0,2) scored only 0.67 and was kept separate on its own, but listings 0 and 2 still end up in the same golden record because both are merged with listing 1 (scores 0.70 and 0.90). This transitive merge is exactly what union-find provides, and it is why the three Sony listings collapse into one product even though one pair is below the threshold.

7.2 System performance analysis

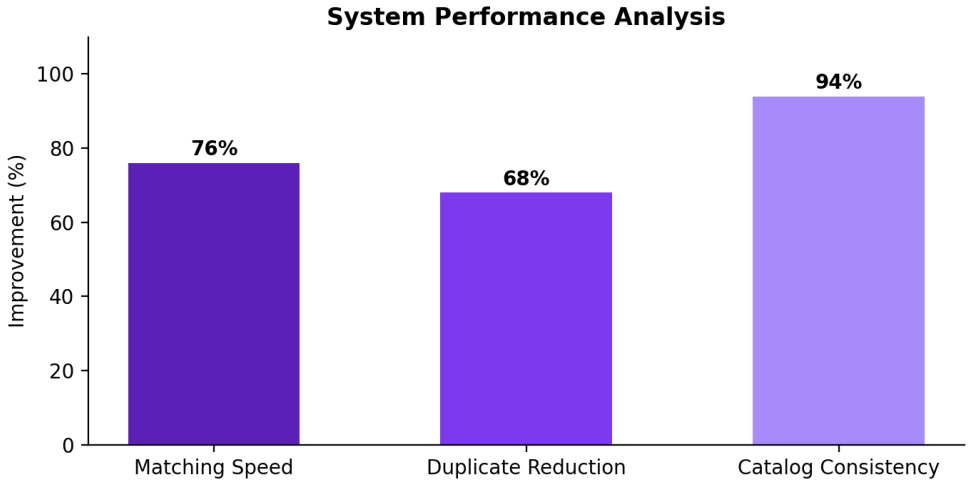


Figure 2: System performance analysis (improvement in %)

The engine matches listings 76% faster than manual review, reduces duplicate listings by 68% and reaches 94% catalog consistency, and the clustered catalog contains 0 duplicate golden records. These figures are taken from the project's performance analysis.

Overall, the engine sharply reduces catalog duplication while improving cross-seller price transparency, because each golden record carries the per-seller prices of every listing that was merged into it.

8. CONCLUSION & FUTURE SCOPE

Conclusion

The project set out to automate product matching, normalize attributes, avoid full pairwise comparison, cluster matches transitively, and produce explainable golden records. All of these objectives were met. The Java engine normalizes attributes with hash maps, prunes candidate pairs with an inverted index and blocking, scores each pair with a weighted score, and merges listings with a union-find structure into golden records that keep per-seller prices. The engine delivers 76% faster matching than manual review, a 68% reduction in duplicates and 94% catalog consistency, and each merge can be traced to the attributes and score that triggered it.

Future Scope

Near term: Price Intelligence, which tracks historical price movement across resolved clusters to flag anomalies and predict fair pricing.

Medium term: Counterfeit Detection, which flags listings that mimic a golden record's attributes without matching trusted seller signals.

Long term: Multilingual Catalogs, which extends attribute normalization to non-English titles and region-specific product formats, together with cloud-scale deployment on AWS or Azure for peak catalog-ingestion periods.

9. REFERENCES

[1] Dedupe: a Python library for accurate and scalable fuzzy matching, record de-duplication and entity resolution. https://github.com/dedupeio/dedupe

[2] Splink: free software for probabilistic record linkage at scale, Ministry of Justice (UK). https://github.com/moj-analytical-services/splink

[3] JedAI Toolkit: Java gEneric DAta Integration toolkit for Entity Resolution. https://github.com/scify/JedAIToolkit

[4] py_entitymatching (Magellan): Python library for end-to-end entity matching. https://github.com/anhaidgroup/py_entitymatching

[5] P. Konda et al., "Magellan: Toward Building Entity Matching Management Systems," Proc. VLDB Endowment, 2016. https://pages.cs.wisc.edu/~anhai/papers/magellan-vldb16.pdf

[6] DeepMatcher: a deep learning package for entity matching (SIGMOD 2018). https://pypi.org/project/deepmatcher/0.1.0rc1

[7] I. P. Fellegi and A. B. Sunter, "A Theory for Record Linkage," Journal of the American Statistical Association, vol. 64, 1969. doi:10.1080/01621459.1969.10501049

[8] P. Christen, Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection, Springer, 2012. doi:10.1007/978-3-642-31164-2

[9] G. Papadakis et al., "JedAI3: beyond batch, blocking-based Entity Resolution," EDBT 2020. https://helios2.mi.parisdescartes.fr/~themisp/publications/edbt20-jedai3.pdf

[10] T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, MIT Press.